



COLLEGE OF ENGINEERING KIDANGOOR
DEPT. OF COMPUTER SCIENCE & ENGINEERING



NBA Accredited
2022-2025

FIRE SPARK: WILDFIRE AND SMOKE DETECTION SYSTEM

GUIDE : Dr. Rekha K.S.

Assistant Professor

Dept. of CSE

College of Engineering Kidangoor

Group Members:

Aparna Reghu (KGR21CS021)

Ciya Joe (KGR21CS030)

Deepa Mariam Thomas (KGR21CS033)

Geethu K. (KGR21CS038)

Marion Punnen Kurien (KGR21CS055)

DEPARTMENT VISION

To become a center of excellence in Computer Science and Engineering imparting quality professional education to develop competent professionals with social values who are capable of life long learning.

DEPARTMENT MISSION

To impart quality technical education to students at undergraduate level through constant knowledge upgradation by maintaining pace with the latest sophisticated innovations, research and development and industry interaction in the field of Computer Science and Engineering with focus on lifelong learning for the well-being of the society.

ABSTRACT

- Wildfires pose a severe environmental threat, causing extensive loss of life, property, and natural resources.
- AI-powered wildfire, smoke, and fog detection system for proactive disaster management.
- Uses CNN with FastAI for high-accuracy detection.
- Trained on diverse datasets for high detection accuracy.
- Real-time detection with automated email alerts for rapid response
- Scalable, deep-learning-based solution for disaster mitigation and proactive management.

CONTENTS

- Introduction
- Existing System
- Proposed System
- Problem Statement
- Literature Review
- System Architecture
- Modules and Functionalities
- Design Diagrams
- System Requirements
- Implementation and Results
- Conclusion
- References



INTRODUCTION

- Wildfires cause severe environmental and infrastructural damage.
- Traditional detection methods are slow, costly, and less accurate.
- AI-powered detection using CNN with FastAI improves efficiency.
- Detects wildfire, smoke, and fog from images/videos in real-time.
- Trained on diverse datasets for high accuracy across conditions.
- Includes severity estimation and instant email alerts for response.
- Automated, scalable, and reliable for disaster management.

EXISTING SYSTEM

Disadvantages of Existing System (Traditional Methods) :-

- Fire Lookouts & Manual Observation
- Ground Sensors
- Satellite Monitoring
- Limited Automation
- High False Alarm Rate
- Weather Dependency
- Resource-Intensive

PROPOSED SYSTEM

Advantages of Proposed System (AI-Powered Detection):-

- Uses CNN with FastAI
- Real-time image & video analysis
- Trained on diverse datasets
- Severity estimation
- Email alerts
- Scalable & automated

PROBLEM STATEMENT

To develop an AI-powered wildfire, smoke and fog detection system that quickly identifies fire hazards using CNNs, processes real-time data from cameras and drones, and sends automated alerts to emergency responders to minimize damage and enhance public safety.

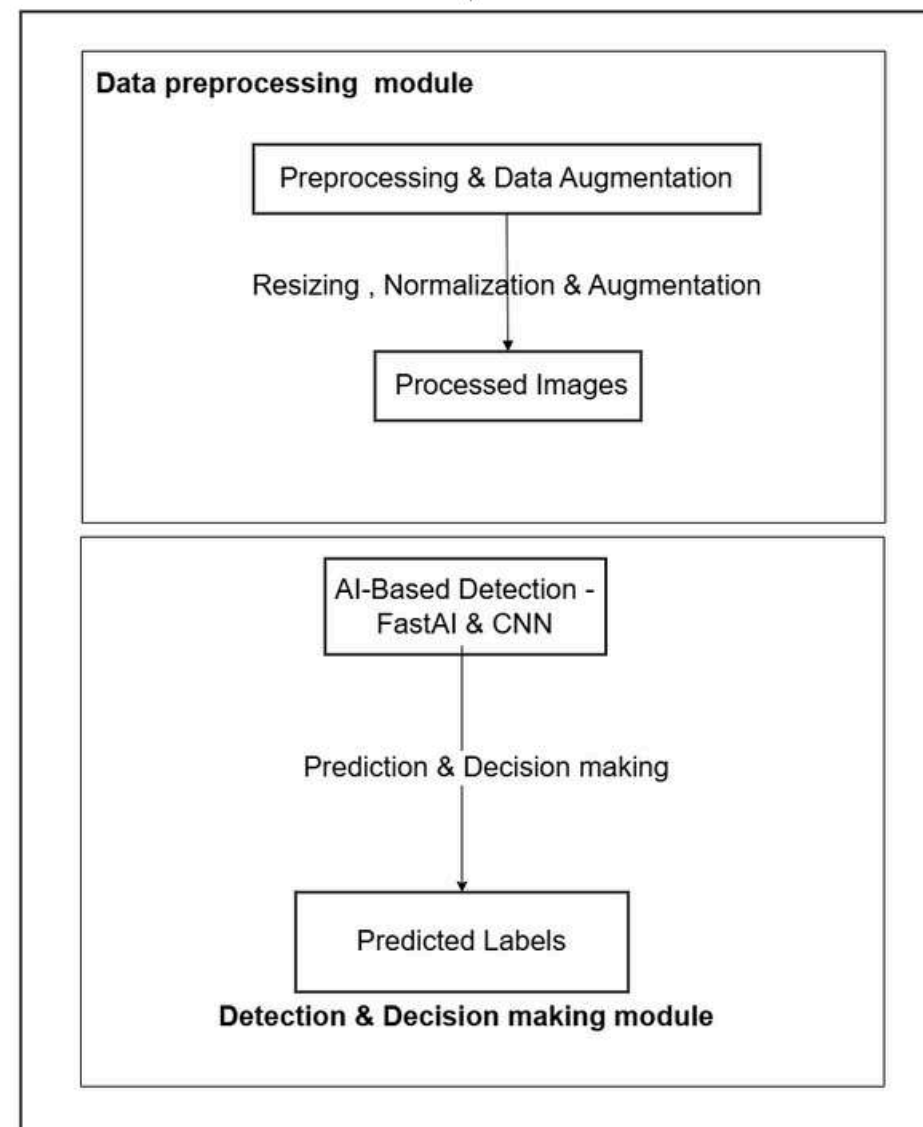
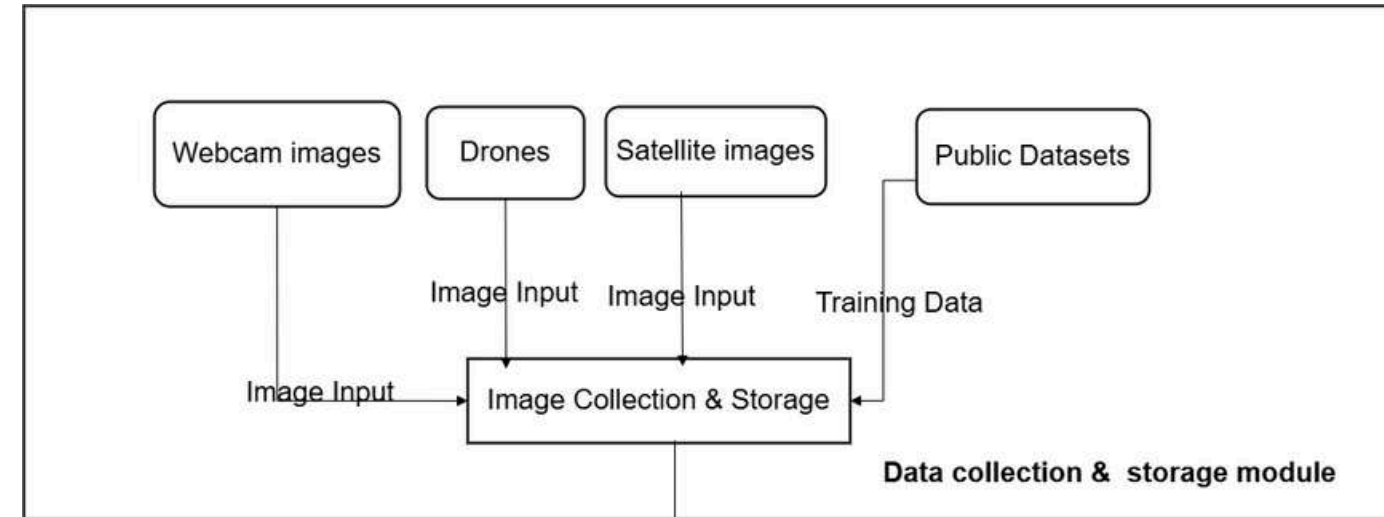


LITERATURE REVIEW

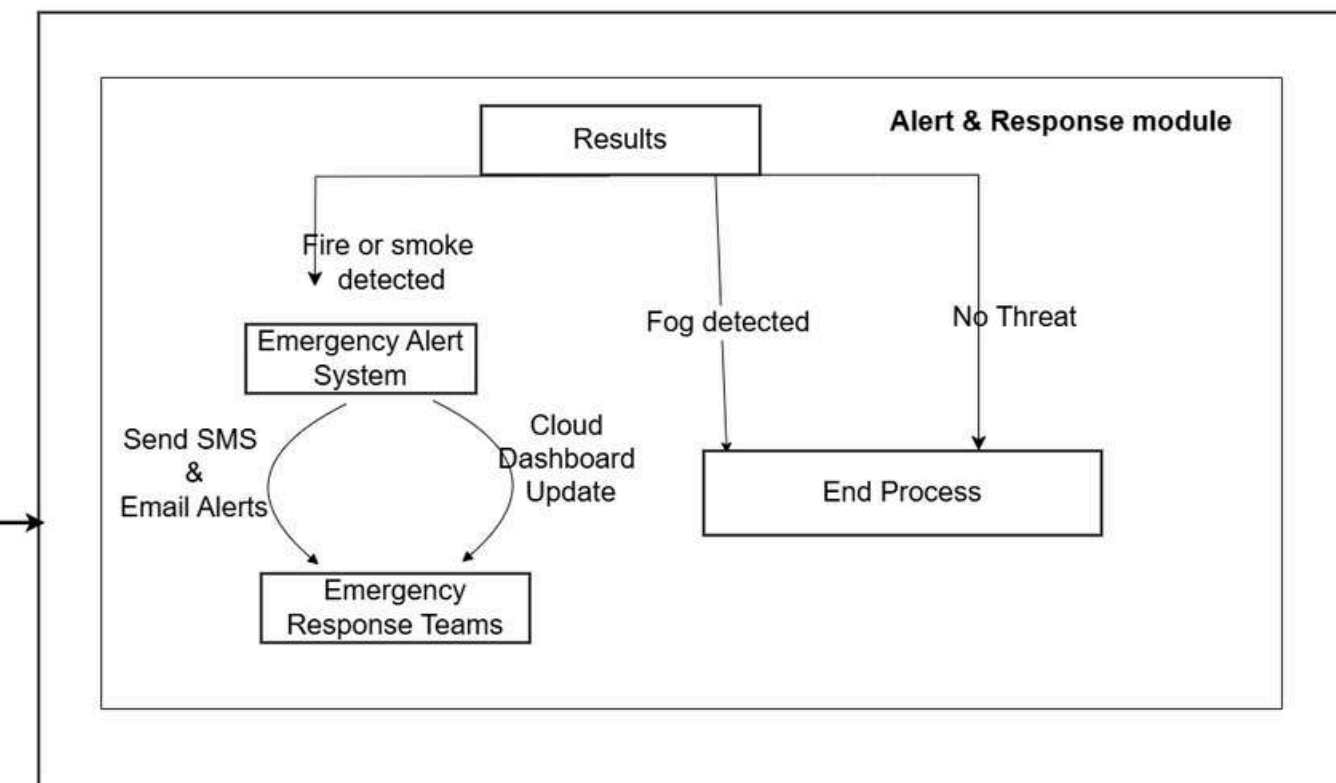
SL NO	AUTHORS	TITLE OF PUBLICATION	DESCRIPTION
1.	Khan Muhammed Salman Khan, Syed Hassan Ahamed	Efficient Fire Detection for Uncertain Surveillance Environment	- Uses ML, image processing, and pattern recognition. - Works in real-time with visual and thermal data.
2.	Afonso M. Gonçalves, Tomás Brandão, João C. Ferreira	Wildfire Detection with Deep Learning	- CNN model detects smoke in complex outdoor images. - Uses real-world sensor data for accuracy.
3.	Arpit Jadon, Mohd. Omama, Akshay Varshney, Rishabh Sharma	<u>FireNet</u> : Optimized for IoT & Real-Time Detection	- CNN model detects smoke in video feeds. - Handles occlusions and lighting variations.
4.	Suleyman Aslan, Ugur G. Udukbay, B. Ugur T. <u>Oreyin</u>	Early Wildfire Smoke Detection Based on DCGAN	- Uses DCGAN for synthetic wildfire smoke detection. - Two-stage training improves real-world performance.
5.	F. M. Amin Hossain, Chi Yuan, Chun-Yi Sun	Wildfire Flame and Smoke Detection Using ANN	- Processes single images without temporal data. - Enhances efficiency using ANN-based classification.

SL NO	AUTHORS	TITLE OF PUBLICATION	DESCRIPTION
6.	Qi-xing Zhang, Gao-Hua Lin, Yong-ming Zhang, Gao Xu	Wildland Forest Fire Smoke Detection Using Faster R-CNN	- Faster R-CNN trained on synthetic and real images. - Eliminates manual feature extraction for detection.
7.	Sebastien Frizzi, Moez Bouchouicha, Jean-Marc Ginoux, Eric Moreau	Convolutional Neural Network for Smoke and Fire Semantic Segmentation	- Uses CNN for pixel-wise fire and smoke detection. - Outperforms traditional segmentation techniques.
8.	Yazan Al-Smadi, Mohammad Alauthman, Ahmad Al-Qerem, Amjad Aldweesh	Early Wildfire Smoke Detection Using YOLO Models	- Compares YOLO versions for real-time smoke detection. - Aims at improving wildfire monitoring accuracy.
9.	Khan Ragib Mahmud, Lingxiao Wang, Xiyuan Liu, J. Liahao, Sunzid Hassan	Deep Learning-Based Wildfire Smoke Detection Using UAVs	- CNN trained on aerial images from drones. - Enables early wildfire monitoring using UAVs.
10.	Ahmed S. Mahdi, Sawсен A. Mahmood	Analysis of Learning Methods for Early Wildfire Detection	- Compares ML techniques: Decision Trees, SVM, CNN, YOLO. - Highlights computational demands of CNN-based models.

SYSTEM ARCHITECTURE



Predicted results



MODULES AND ITS FUNCTIONALITIES

MODULE NAME	DESCRIPTION
Data Collection and Storage Module	Collects and stores wildfire-related image data from various sources, ensuring proper dataset management.
Data Preprocessing Module	Processes raw data by resizing, normalizing, and augmenting images for model training, ensuring consistency.
Detection and Decision-Making Module	Utilizes deep learning models to classify images as wildfire, smoke, fog, or normal conditions.
Alert and Response Module	Generates real-time alerts based on model predictions, sending notifications to relevant authorities.

Data Collection & Storage Module

COMPONENT NAME	DESCRIPTION
Image Sources	Captures images from satellites, drones, and surveillance cameras.
Dataset Management	Organizes collected images into labeled categories.
Cloud/Local Storage	Stores datasets for further processing and model training.
Data Retrieval	Fetches stored images for preprocessing and analysis

Data Preprocessing Module

COMPONENT NAME	DESCRIPTION
Image Resizing	Ensures uniform image dimensions for model training.
Normalization	Scales pixel values to enhance model accuracy.
Data Augmentation	Applies transformations (rotation, flipping) to increase dataset variability.
Noise Reduction	Filters unwanted distortions to improve detection accuracy.

Detection & Decision Making Module

COMPONENT NAME	DESCRIPTION
Feature Extraction	Identifies key features in images for classification.
Deep Learning Model	Classifies images into wildfire, smoke, fog, or normal categories.
Confidence Score	Assigns probability values to predictions for accuracy assessment.
Model Optimization	Fine-tunes model performance using training feedback.

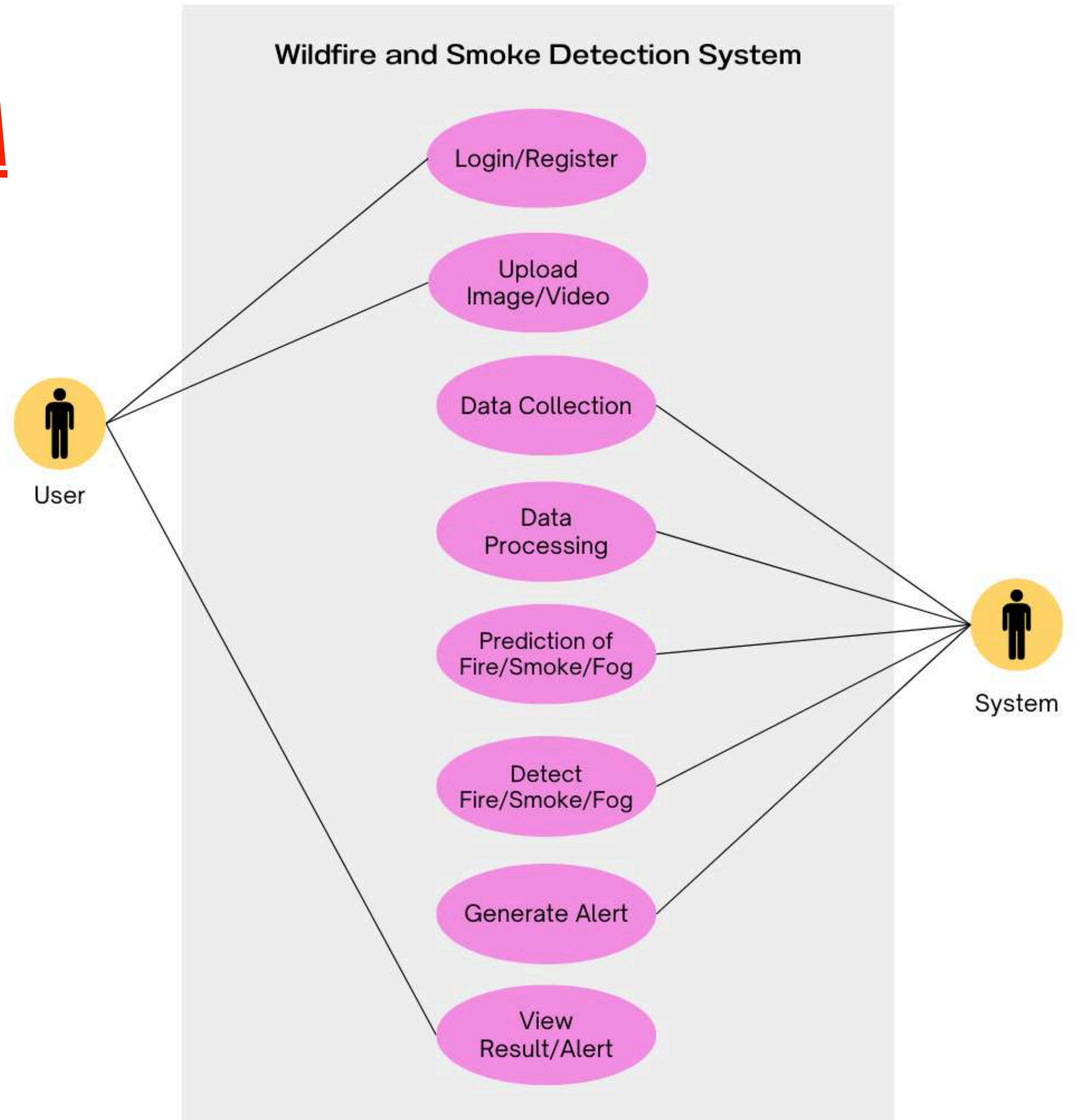
Alert & Response Module

COMPONENT NAME	DESCRIPTION
Prediction Analysis	Interprets model results to detect wildfire occurrences.
Alert Trigger System	Sends real-time notifications based on detection results.
Communication Channels	Uses SMS, email, or app notifications to inform authorities.
Logging System	Records detection history for analysis and improvements.

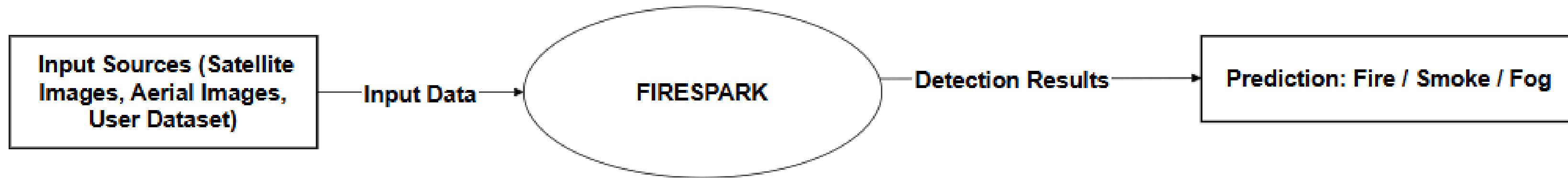


DESIGN DIAGRAMS

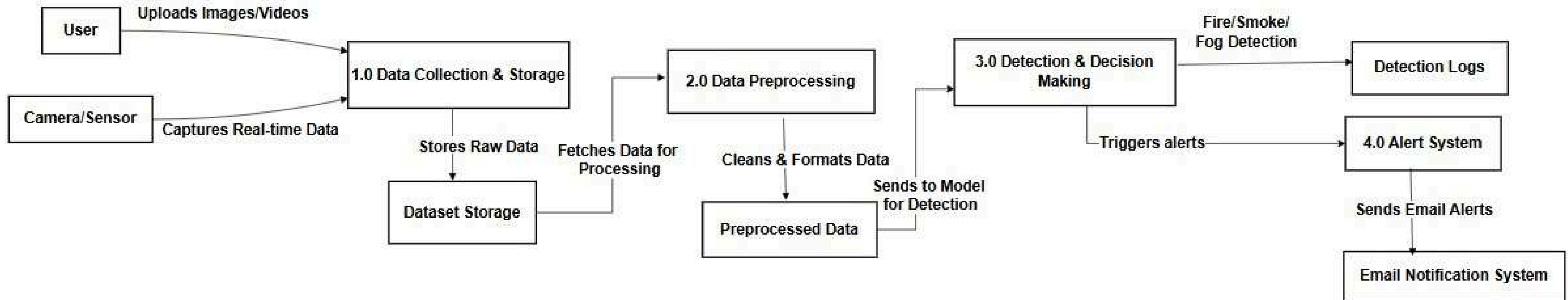
USECASE DIAGRAM



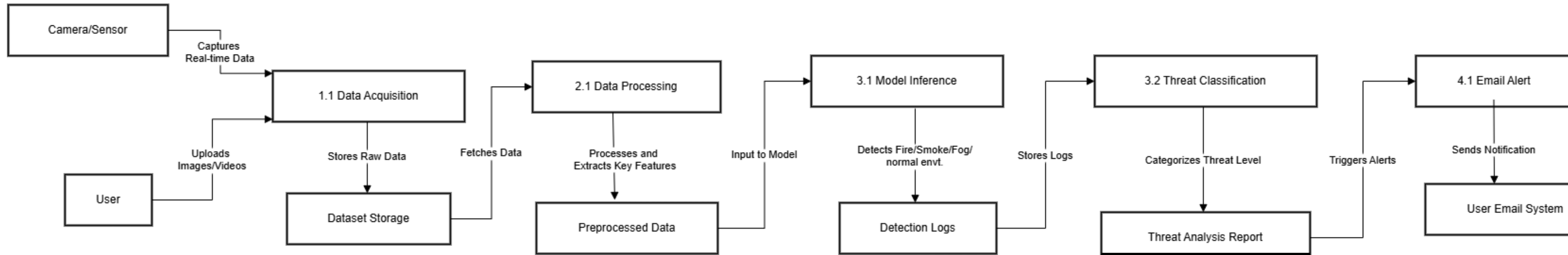
DFD LEVEL - 0



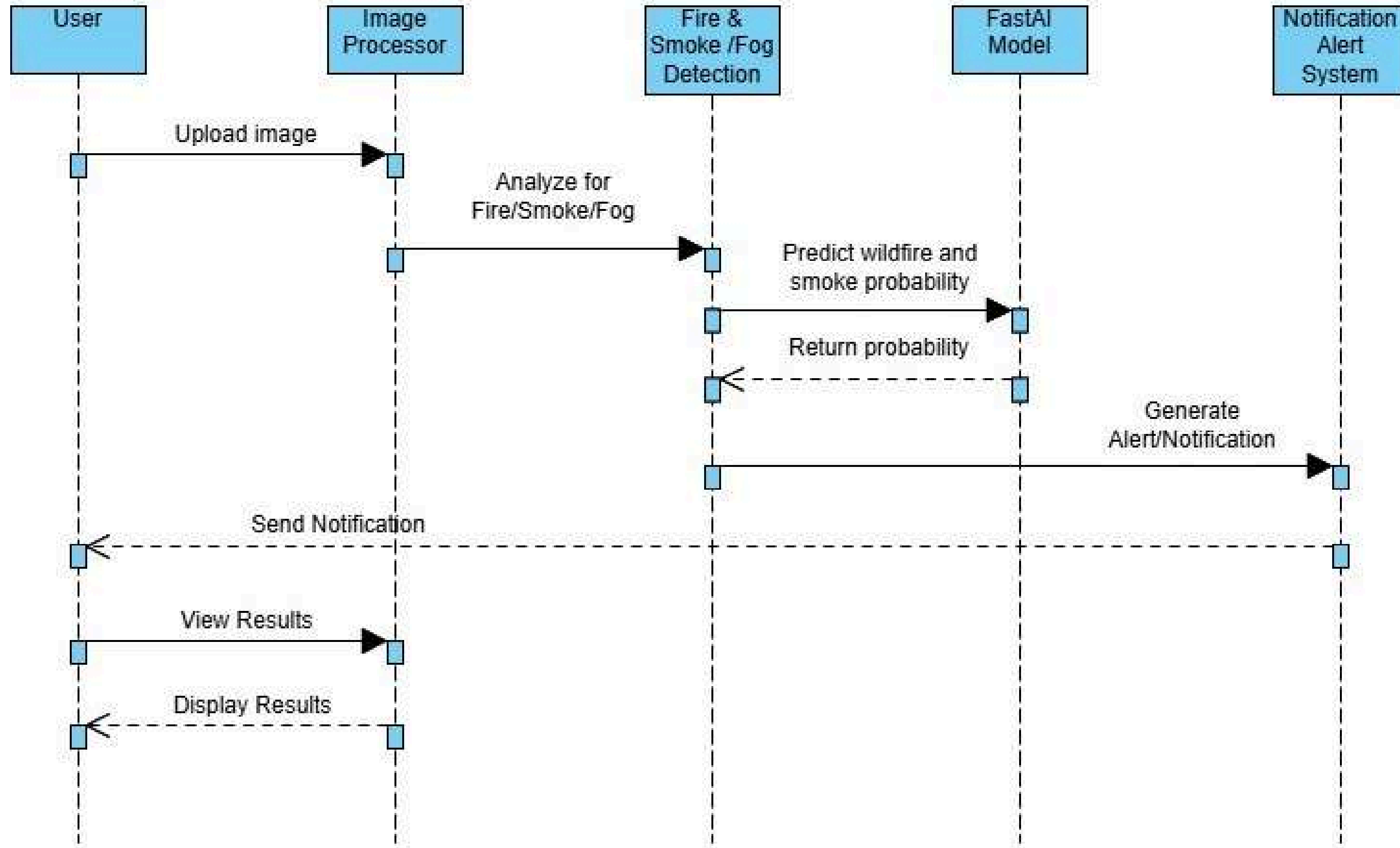
DFD LEVEL - 1



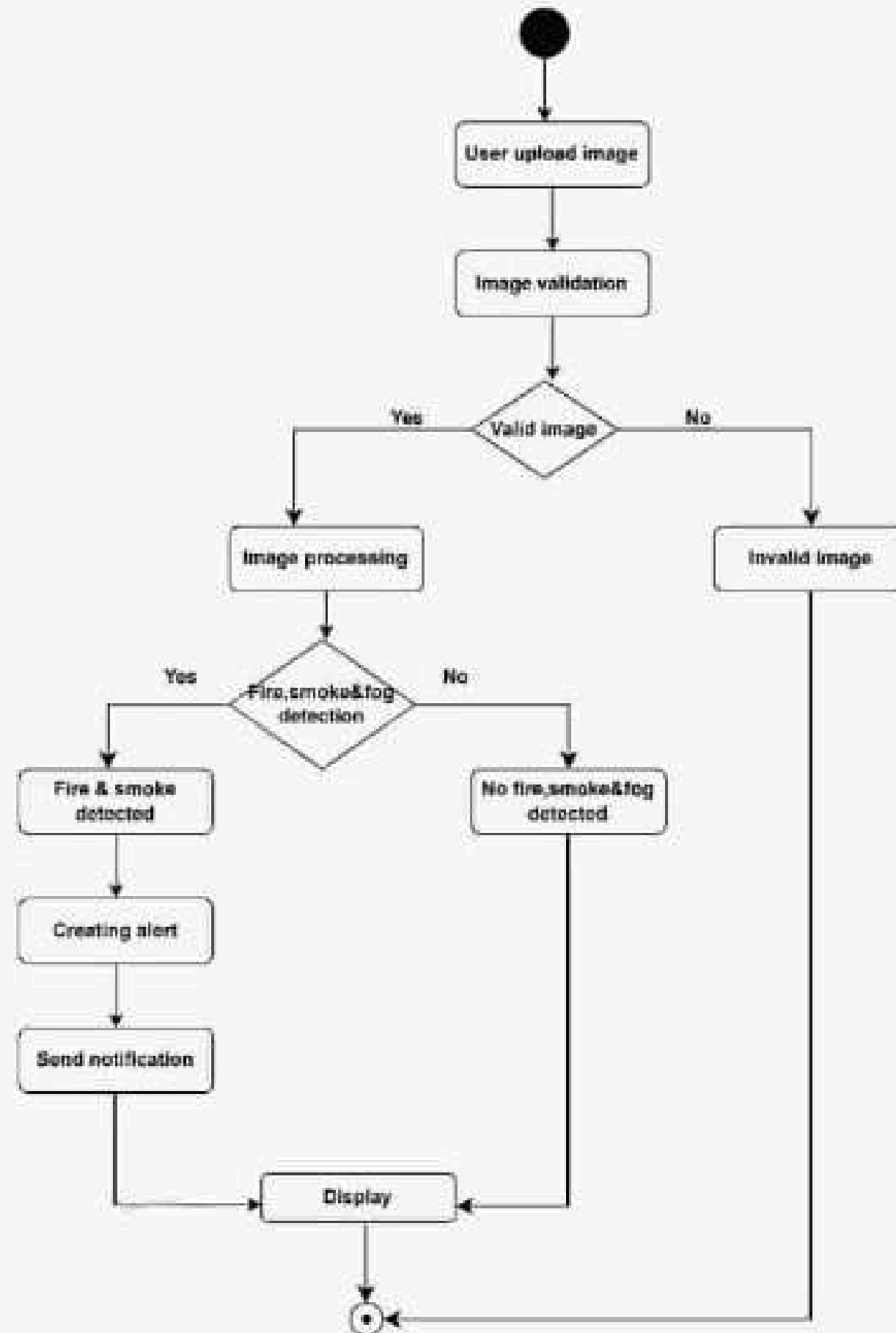
DFD LEVEL - 2



SEQUENCE DIAGRAM



ACTIVITY DIAGRAM



SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS

- ✓ CPU : Intel core I3 Processors
- ✓ Hard Disk Space : 250 GB
- ✓ RAM : 4GB or Above
- ✓ Storage : 512GB and above
- ✓ Clock Speed : 2.6 GHz

SOFTWARE REQUIREMENTS

- ✓ Operating System : Windows 10 and above
- ✓ Front End :
HTML,CSS,JAVASCRIPT,BOOTSTRAP
- ✓ Backend : Python Django, Fast AI Technology
- ✓ IDE : Pycharm
- ✓ Database : SQLite3

IMPLEMENTATION & RESULTS

```
[4]: path = Path(r"D:\firespark") # Update with your actual dataset path
dls = ImageDataLoaders.from_folder(path, train='.', valid_pct=0.2, seed=42, item_tfms=Resize(224))
dls.show_batch(max_n=20, figsize=(6,6))
```



Images in our dataset

IMPLEMENTATION & RESULTS

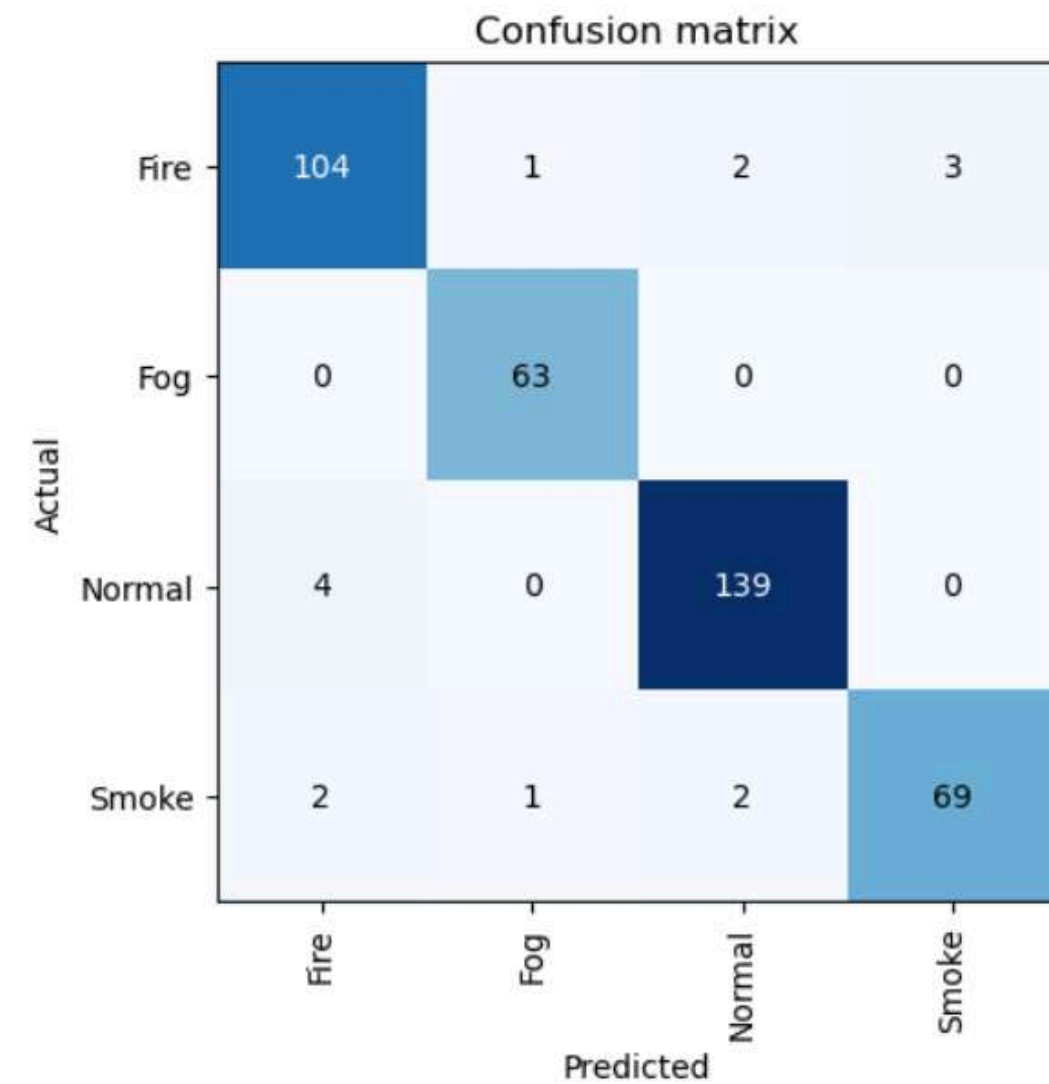
```
[7]: model = vision_learner(dls, resnet34, metrics=accuracy)
model.fine_tune(5)  # Adjust number of epochs based on performance
```

epoch	train_loss	valid_loss	accuracy	time
0	1.312640	0.359814	0.894872	05:10
epoch	train_loss	valid_loss	accuracy	time
0	0.391985	0.171635	0.933333	06:12
1	0.218180	0.126245	0.956410	07:36
2	0.137307	0.145678	0.961538	09:37
3	0.088115	0.116467	0.964103	08:11
4	0.056741	0.114172	0.966667	11:31

Epoch Training progress of our Fast AI model

IMPLEMENTATION & RESULTS

```
prediction.plot_top_losses(5, figsize=(8,6))
```



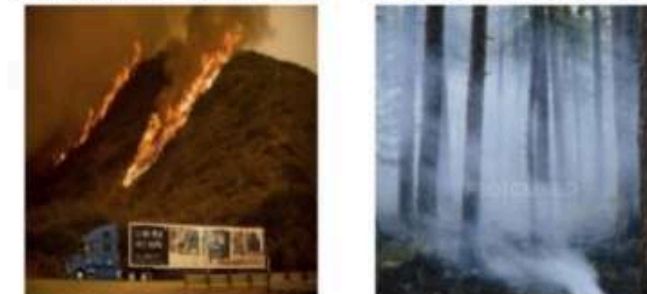
Confusion matrix

Prediction/Actual/Loss/Probability

Smoke/Fire / 5.73 / 1.00 Normal/Smoke / 5.55 / 1.00 Fire/Normal / 4.51 / 0.99

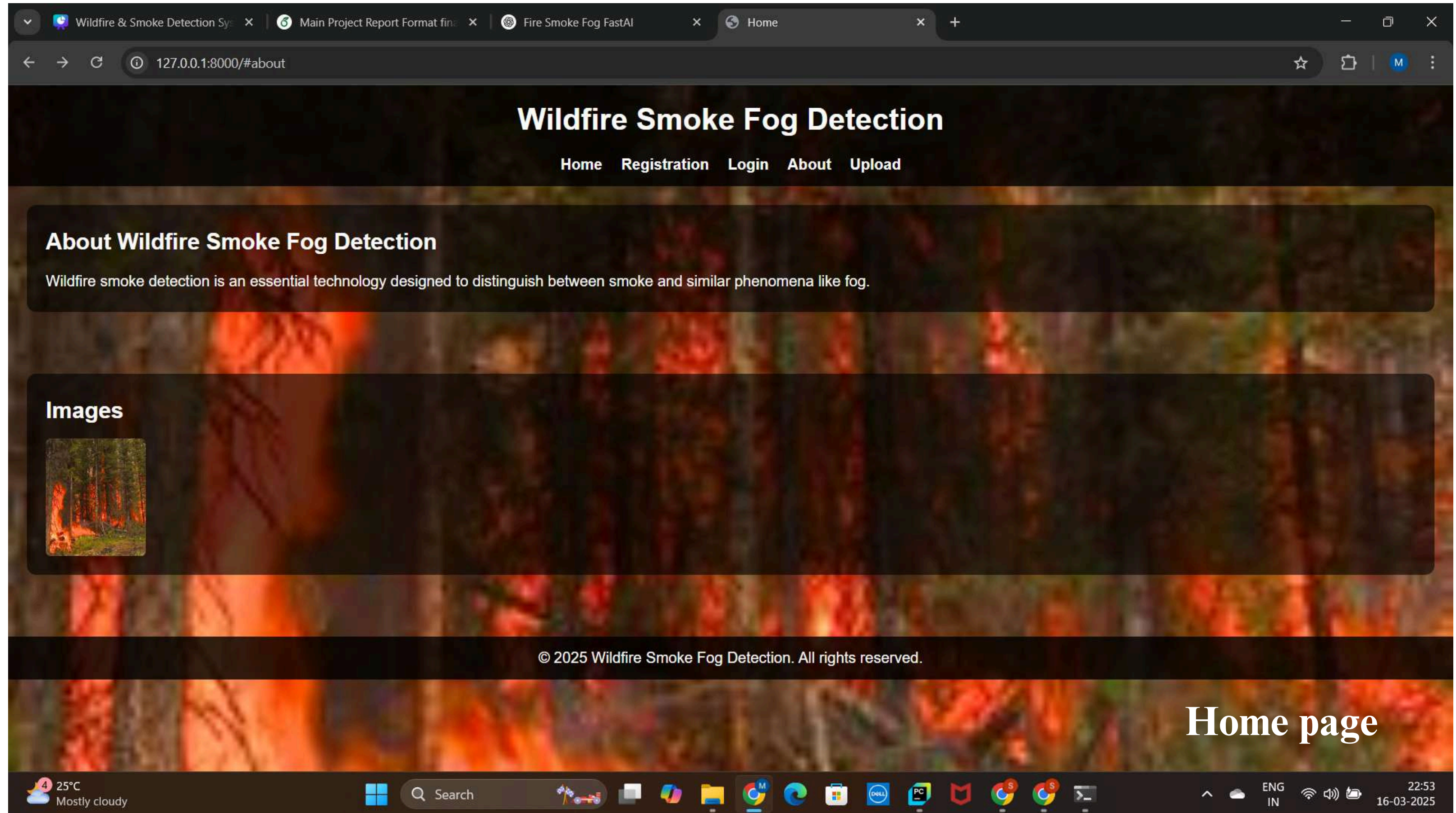


Normal/Fire / 3.86 / 0.97 Fog/Smoke / 3.22 / 0.96



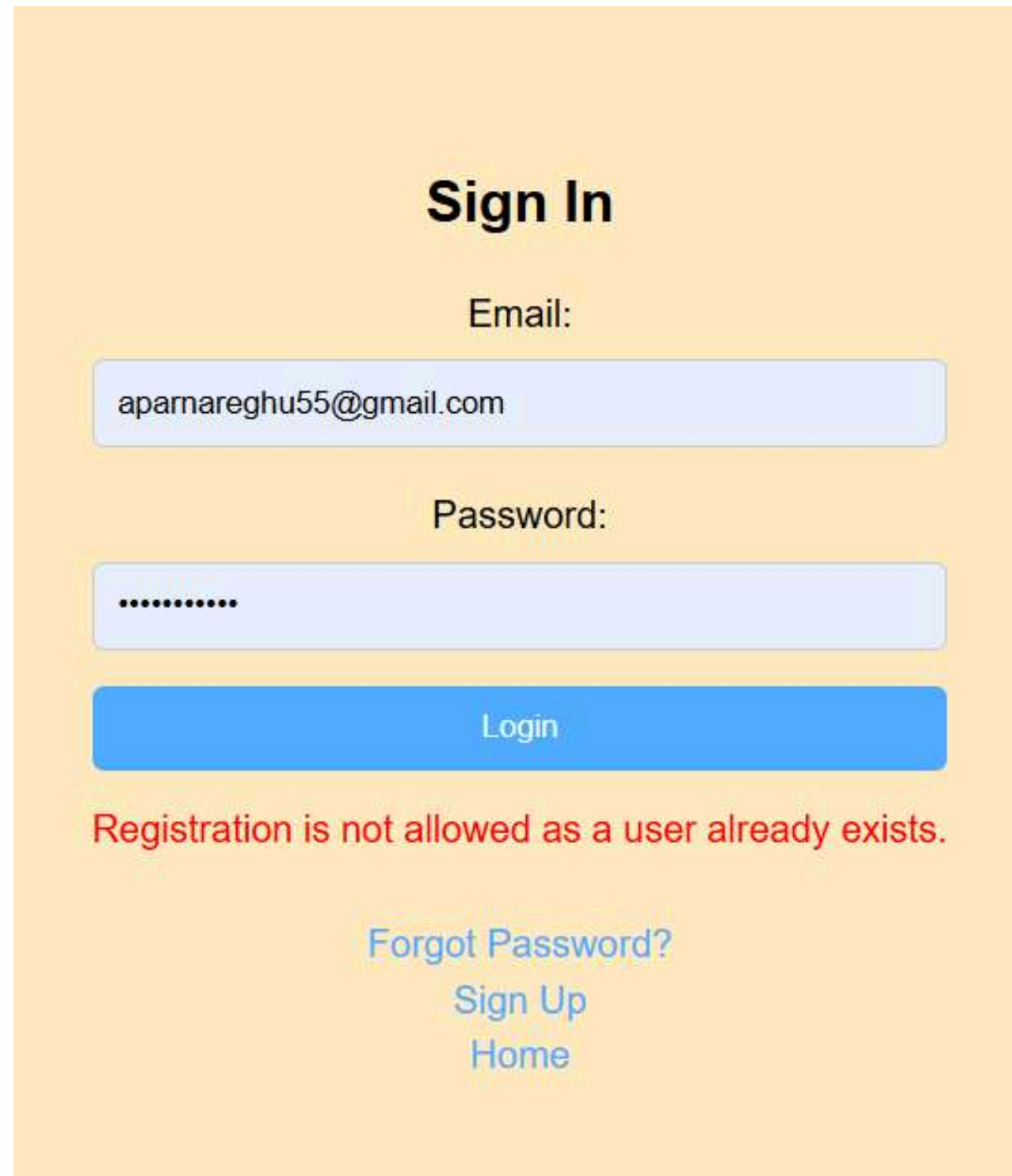
Prediction v/s actual classification

IMPLEMENTATION & RESULTS



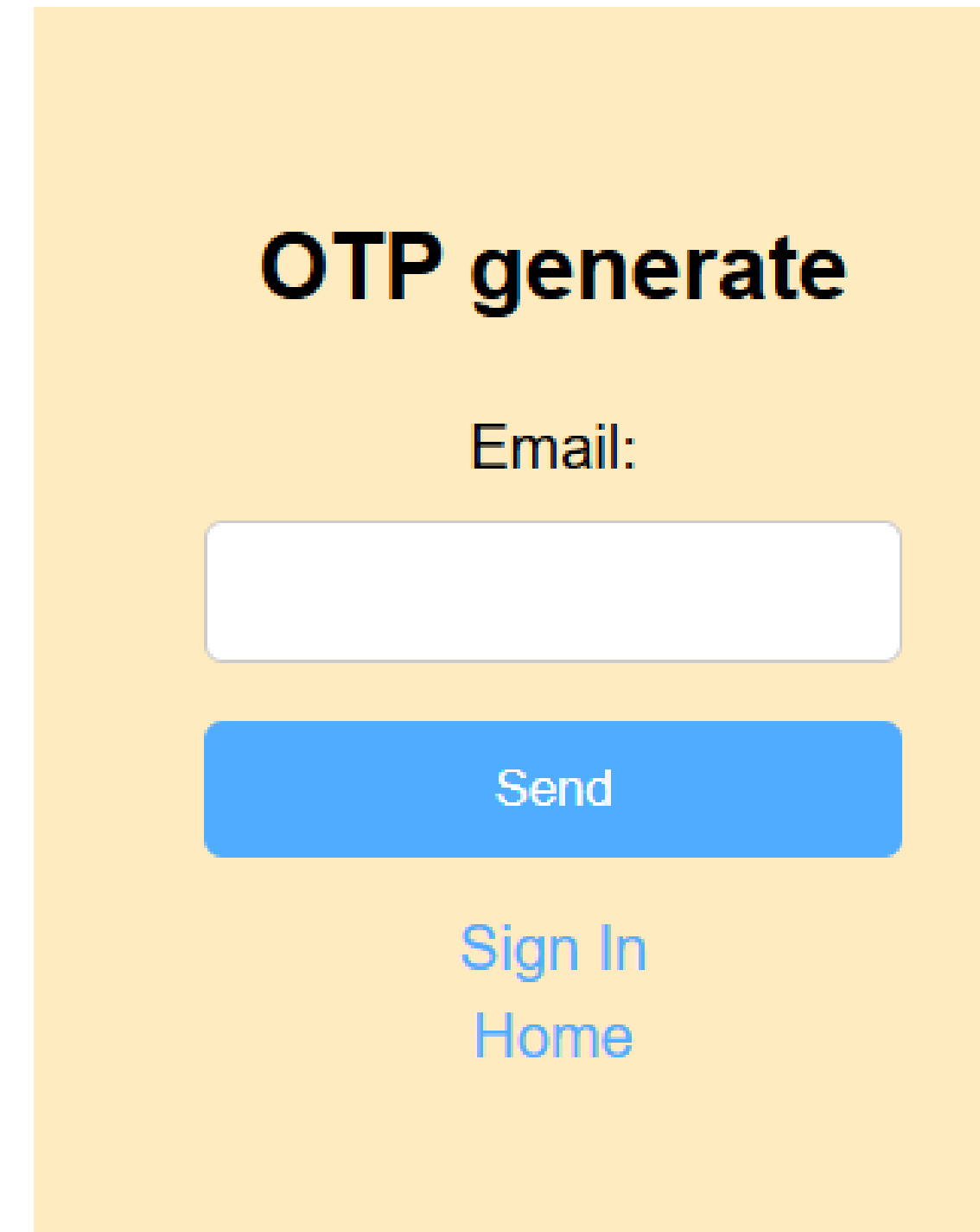
Home page

IMPLEMENTATION & RESULTS



The screenshot shows a 'Sign In' form on a light orange background. It includes a title 'Sign In', an 'Email:' label with a text input field containing 'aparnareghu55@gmail.com', a 'Password:' label with a password input field showing dots, and a blue 'Login' button. Below the button is a red error message: 'Registration is not allowed as a user already exists.' At the bottom, there are links for 'Forgot Password?', 'Sign Up', and 'Home'.

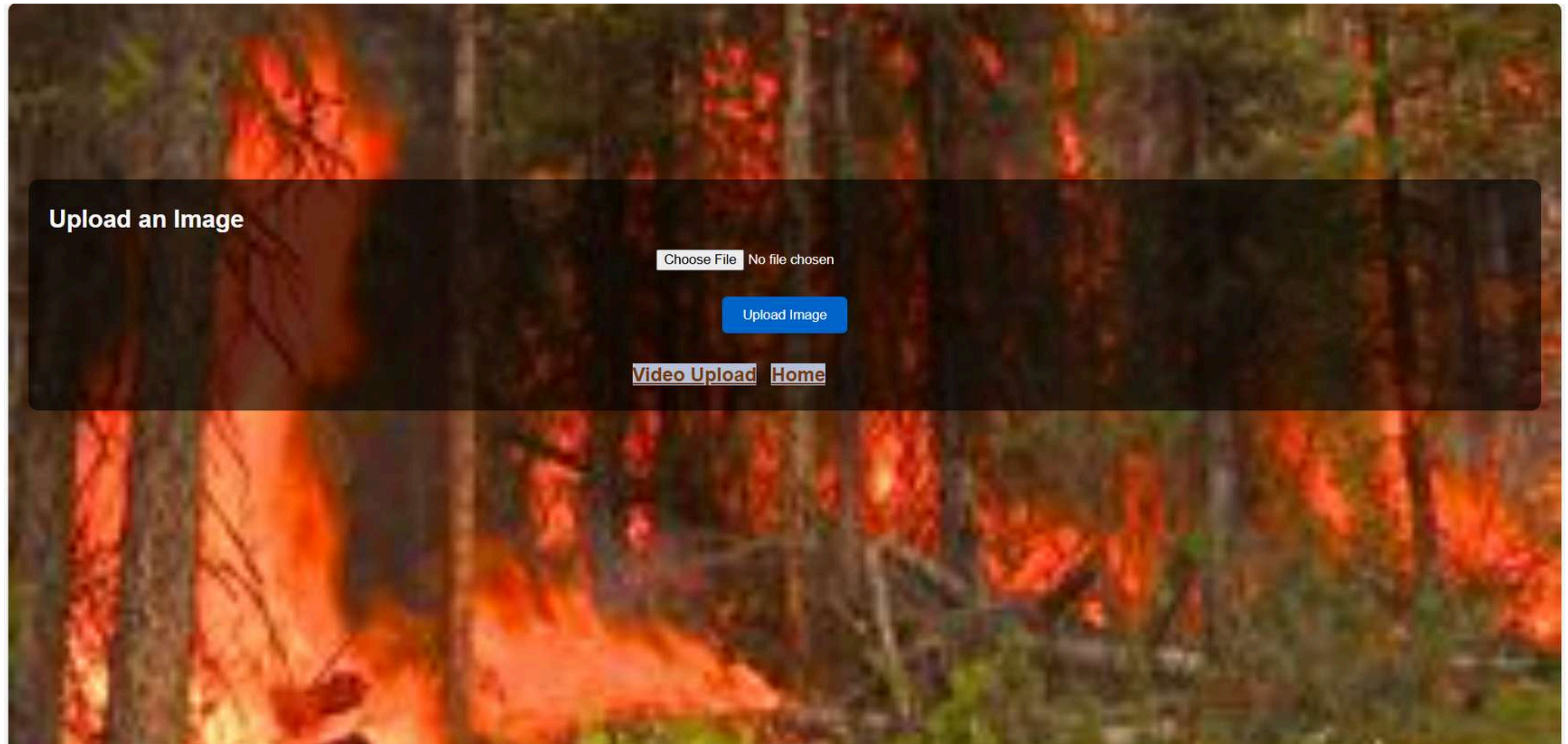
Sign in page (Admin)



The screenshot shows an 'OTP generate' form on a light orange background. It includes a title 'OTP generate', an 'Email:' label with an empty text input field, and a blue 'Send' button. Below the button are links for 'Sign In' and 'Home'.

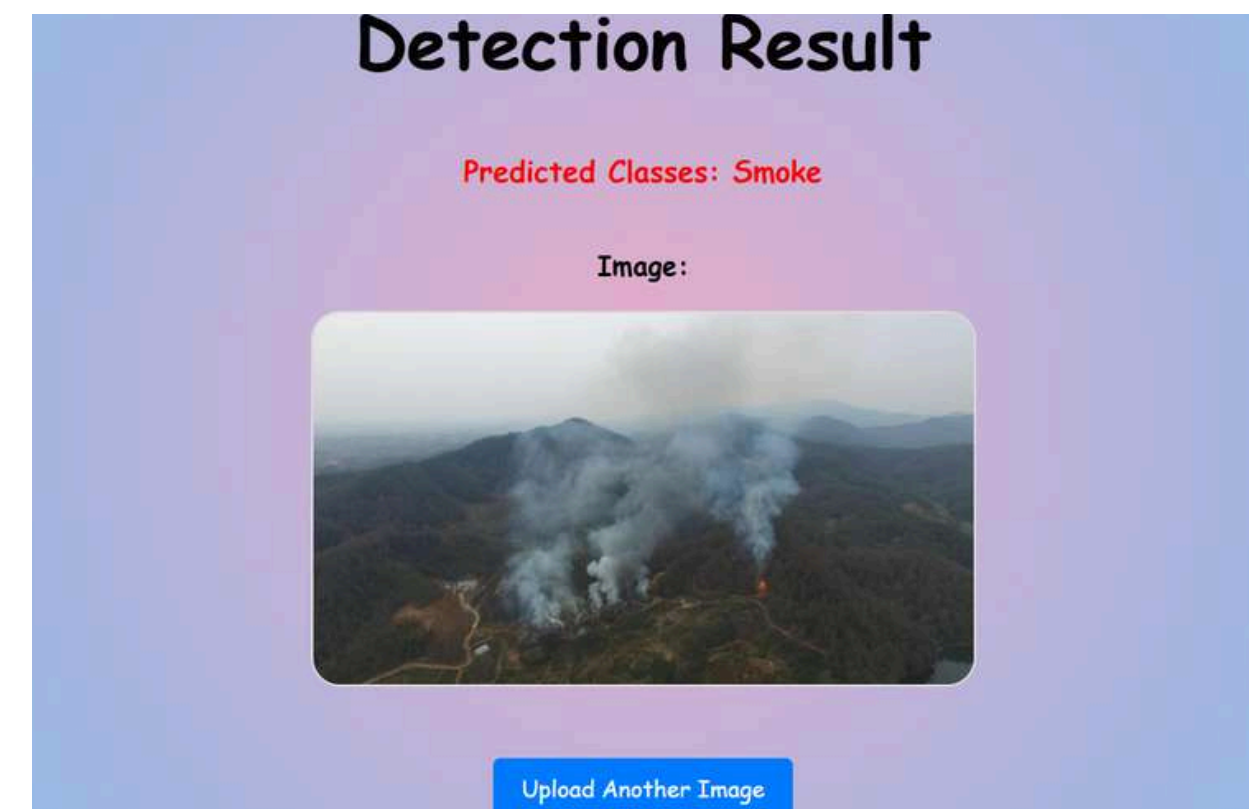
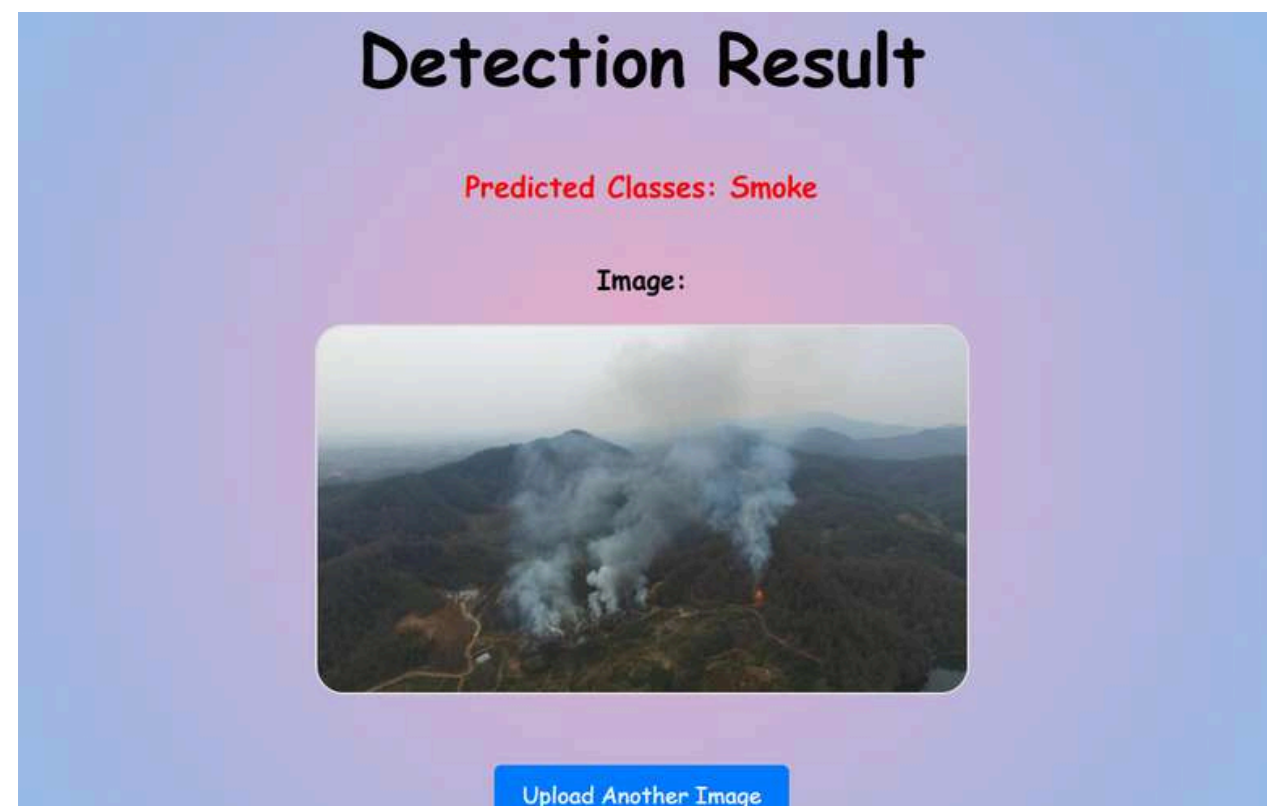
Otp verification page

IMPLEMENTATION & RESULTS



Upload image page

IMPLEMENTATION & RESULTS



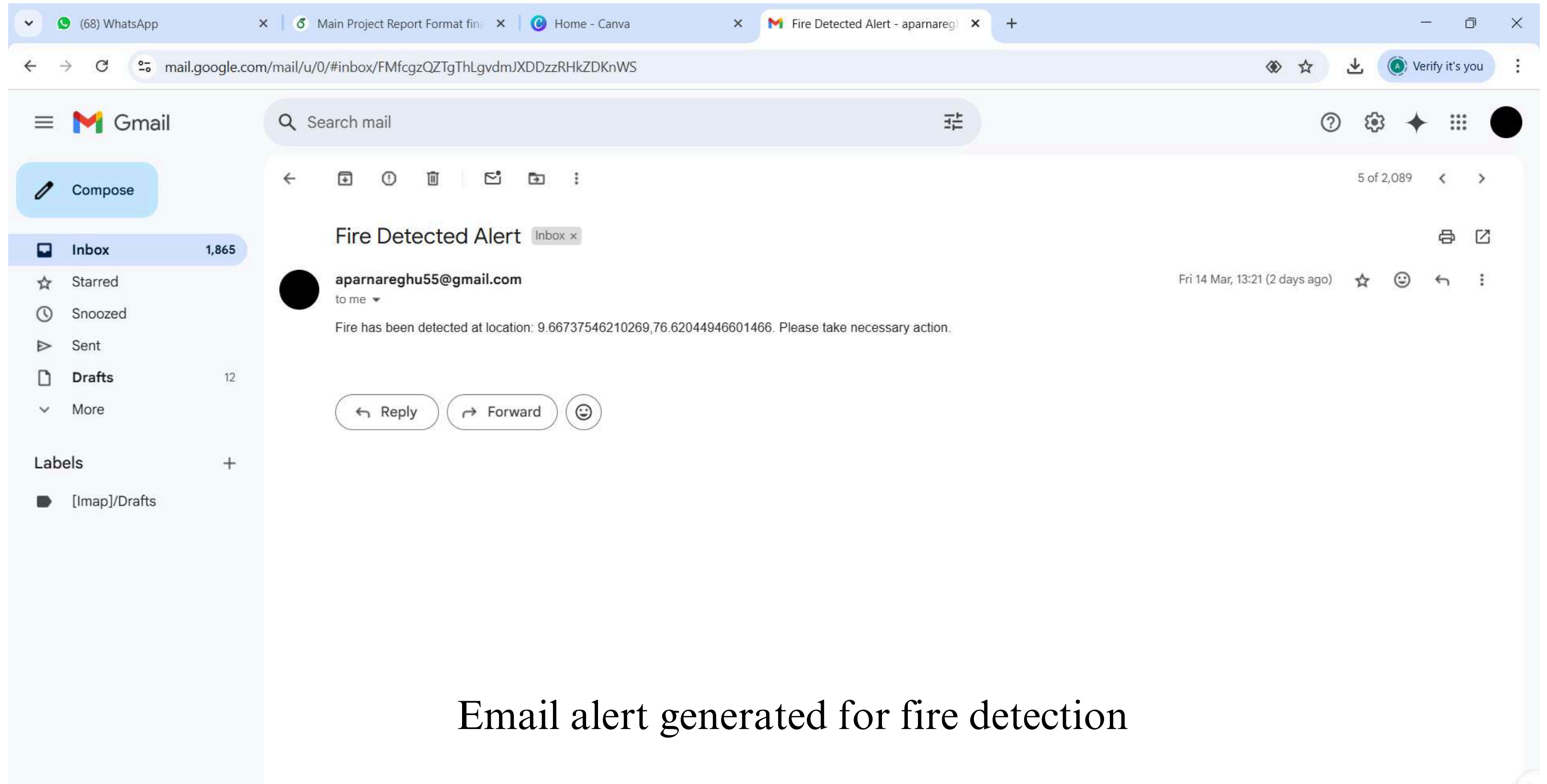
Detection of fire, smoke, fog & normal environment

IMPLEMENTATION & RESULTS

Prediction Result					Logout
ID	Location	Prediction	Date	Time	
1	10.003045939904885,76.30588909576308	Fire	2025-03-13	15:56:44	
2	10.003049679392998,76.30589226417497	Normal	2025-03-13	16:08:05	
3	10.003007757800903,76.30592868171644	Normal	2025-03-13	16:09:03	
4	10.003024538492975,76.30590638673475	Normal	2025-03-13	16:10:15	
5	10.1837,76.3887	Fog	2025-03-14	07:14:23	
6	10.1837,76.3887	Fog	2025-03-14	07:15:26	
7	10.1837,76.3887	Normal	2025-03-14	07:16:34	
8	10.1837,76.3887	Normal	2025-03-14	07:16:56	
9	10.1837,76.3887	Smoke	2025-03-14	07:17:24	
10	10.1837,76.3887	Smoke	2025-03-14	07:17:24	
11	10.1837,76.3887	Smoke	2025-03-14	07:17:25	
12	10.1837,76.3887	Fire	2025-03-14	07:18:53	
13	10.1837,76.3887	Fire	2025-03-14	07:20:35	
14	10.1837,76.3887	Fire	2025-03-14	07:21:51	

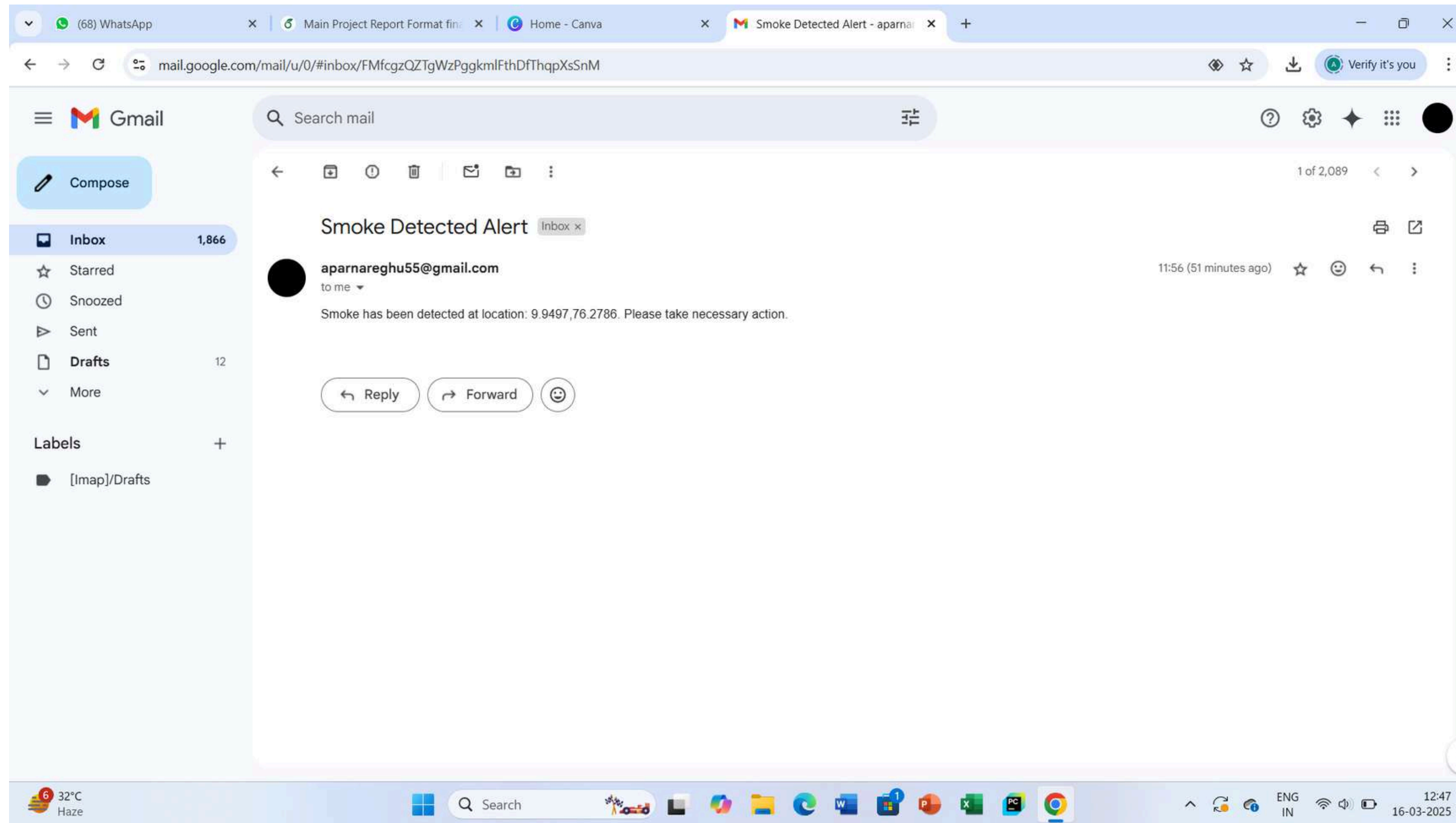
Detection log

IMPLEMENTATION & RESULTS



Email alert generated for fire detection

IMPLEMENTATION & RESULTS



Email alert generated for smoke detection

PUBLICATION DETAILS

- **Paper Title: Innovations in Wildfire and Smoke Detection**
- **Status : Published**
- **Authors : Aparna Reghu, Ciya Joe Deepa Mariam Thomas, Geethu K., Marion Punnen Kurien, Rekha K.S.**
- **Journal Name: International Journal of Scientific Research in Engineering and**
- **Management (IJSREM)**
- **Publication Month: February Edition**

- **Reference:**

[1]. [Aparna Reghu, Ciya Joe, Deepa Mariam Thomas, Geethu K., Marion Punnen Kurien, Rekha .S.], "Innovations in Wildfire and Smoke Detection: A Comprehensive Survey," International Journal of Scientific Research in Engineering and Management (IJSREM), vol. 9, Feb. 2025. DOI: [10.55041/IJSREM41825].

CONCLUSION

- Wildfire, Smoke, and Fog Detection System effectively addresses the limitations of traditional methods by utilizing deep learning for accurate classification.
- The system automates detection and minimizes false alarms by distinguishing between smoke and fog, ensuring faster emergency response.
- It integrates real-time data collection, preprocessing, and decision-making to enhance efficiency and accuracy.
- The alert system ensures that authorities and users receive timely notifications, helping in preventing and controlling wildfires.
- Future improvements include expanding datasets, optimizing model performance, and integrating IoT for enhanced real-time monitoring.

REFERENCES

- [1] Goncalves, A. M., Brando, T., Ferreira, J. C. (2024). Wildfire Detection with Deep Learning
• A Case Study for the CICLOPE Project. IEEE Access.
- [2] Jadon, A., Omama, M., Varshney, A., Ansari, M. S., Sharma, R. (2019). FireNet: a light weight specialized fire smoke detection model for real-time IoT applications. arXiv preprint arXiv:1905.11922.
- [3] Muhammad, K., Khan, S., Elhoseny, M., Ahmed, S. H., Baik, S. W. (2019). Efficient fire detection for an uncertain surveillance environment. IEEE Transactions on Industrial Informatics, 15(5), 3113-3122.
- [4] Aslan, S., Gudukbay, U., Toreyin, B. U., Cetin, A. E. (2019, May). Early detection of wildfire smoke based on motion-based geometric image transformation and deep convolutional generative adversarial networks. In ICASSP 2019- 2019 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) (pp. 8315-8319). IEEE.
- [5] Hossain, F. A., Zhang, Y., Yuan, C., Su, C. Y. (2019, July). Fire and smoke detection using static image features and an artificial neural network. In 2019 1st international conference on industrial artificial intelligence (iai) (pp. 1-6). IEEE.
- [6] Zhang, Q. X., Lin, G. H., Zhang, Y. M., Xu, G., Wang, J. J. (2018). Wildland forest fire smoke detection based on faster R-CNN using synthetic smoke images. Procedia engineering, 211, 441-446.

REFERENCES

- [7] Mahdi, A. S., Mahmood, S. A. (2022, May). Analysis of Deep Learning Methods for Early Wildfire Detection Systems. 2022 5th International Conference on Engineering Technology and its Applications (IICETA) (pp. 271-276). IEEE.
- [8] Frizzi, S., Bouchouicha, M., Ginoux, J. M., Moreau, E. Sayadi, M. (2021). Convolutional neural network for the semantic segmentation of smoke and fire. IET Image Processing, 15(3), 634-647.
- [9] Al-Smadi, Y., Alauthman, M., Al-Qerem, A., Aldweesh, A., Quaddoura, R., Aburub, F., ... Alhmiedat, T. (2023). Early detection of wildfire smoke using different yolo models. Machines, 11(2), 246.
- [10] Mahmud, K. R., Wang, L., Liu, X., Jiahao, L., Hassan, S. (2024, June). Deep Learning-Based Wildfire Smoke Detection Using Uncrewed Aircraft System Imagery. In 2024 21st International Conference on Ubiquitous Robots (UR) (pp. 580-587). IEEE.

**THANK
YOU!**

